

An elementary explicit positivity range for the Li coefficients

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Abstract

Let T be a height to which the Riemann Hypothesis has been rigorously verified by interval arithmetic. We prove by elementary means that the Li coefficients satisfy $\lambda_n \geq 0$ for every integer n with $2 \leq n \leq 2T - O(1)$. For the verified height $T = 3.000175 \cdot 10^{12}$ this gives n up to approximately $6.00035 \cdot 10^{12}$, about $6 \cdot 10^7$ times the range reached by direct computation of λ_n in the literature. The proof uses no information about the location of zeros above T : only the non-negativity of the on-line terms, a window bound for the phases, the classical explicit zero-counting estimate, and an elementary bound for the off-line contribution. No hypothesis is assumed.

1 Introduction

The Li criterion [1] states that the Riemann Hypothesis (RH) is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$, where

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right], \quad (1)$$

the sum being over all non-trivial zeros ρ of $\zeta(s)$, counted with multiplicity, in conjugate pairs (so that $\lambda_n \in \mathbb{R}$). Direct numerical verification of $\lambda_n \geq 0$ has been carried out up to $n \leq 10^5$ (see [2, 3] and the references therein). The purpose of this note is to show that, using only a *verified height* T as input, one obtains an explicit range $n \leq 2T - O(1)$ by elementary estimates, and that no information about the zeros above T is needed beyond a bound on their worst-case contribution.

Throughout, T is a height such that every non-trivial zero with $0 < \gamma \leq T$ has been rigorously verified to lie on the critical line; this is known for $T = 3.000175\,332\,800$ by [5]. We write $N(T) = \#\{\rho : 0 < \gamma \leq T\}$ for the zero-counting function, $F(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}$ and $B_T = \frac{1}{2} \sum_{\gamma > T} \gamma^{-2}$.

2 The phase of an on-line zero

For a zero on the critical line, $\rho = \frac{1}{2} + i\gamma$,

$$1 - \frac{1}{\rho} = \frac{\gamma^2 - \frac{1}{4} + i\gamma}{\gamma^2 + \frac{1}{4}}, \quad \text{so} \quad \left| 1 - \frac{1}{\rho} \right| = 1, \quad (2)$$

and therefore $1 - (1 - 1/\rho)^n = 1 - e^{in\theta_\gamma}$ with

$$\theta_\gamma := \arg\left(1 - \frac{1}{\rho}\right) = \arctan \frac{\gamma}{\gamma^2 - \frac{1}{4}}, \quad \text{hence} \quad \Re[1 - (1 - 1/\rho)^n] = 1 - \cos(n\theta_\gamma) \in [0, 2]. \quad (3)$$

For a zero off the line, write $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$; then

$$\left|1 - \frac{1}{\rho}\right| = e^{c_\rho}, \quad c_\rho = \frac{1}{2} \log \left(1 + \frac{1 - 2\beta}{\beta^2 + \gamma^2}\right). \quad (4)$$

3 Lemma 1: the window bound

Lemma 1. *Let $n \geq 1$ and $\gamma \in [n/2, \min(2n, T)]$. Then*

$$1 - \cos(n\theta_\gamma) \geq C_1(n) := 1 - \cos\left(\frac{1}{2} - \frac{1}{96n^2}\right). \quad (5)$$

Proof. Put $u(\gamma) = \gamma/(\gamma^2 - \frac{1}{4})$ for $\gamma > \frac{1}{2}$, so that $\theta_\gamma = \arctan u(\gamma)$ and

$$u'(\gamma) = \frac{(\gamma^2 - \frac{1}{4}) - 2\gamma^2}{(\gamma^2 - \frac{1}{4})^2} = -\frac{\gamma^2 + \frac{1}{4}}{(\gamma^2 - \frac{1}{4})^2} < 0.$$

Hence $\theta'_\gamma = u'/(1 + u^2) < 0$: θ_γ is strictly decreasing.

For the expansion, $u = \frac{1}{\gamma}(1 + \frac{1}{4\gamma^2} + \frac{1}{16\gamma^4} + O(\gamma^{-6}))$ and $\arctan u = u - \frac{u^3}{3} + \frac{u^5}{5} + O(u^7)$ give

$$\theta_\gamma = \frac{1}{\gamma} - \frac{1}{12\gamma^3} + \frac{1}{80\gamma^5} + O(\gamma^{-7}). \quad (6)$$

Consequently, by monotonicity and $\min(2n, T) \leq 2n$,

$$n\theta_\gamma \geq n\theta_{2n} = \frac{1}{2} - \frac{1}{96n^2} + \frac{1}{2560n^4} + O(n^{-6}) \geq \frac{1}{2} - \frac{1}{96n^2},$$

and likewise $n\theta_\gamma \leq n\theta_{n/2} = 2 - \frac{2}{3n^2} + O(n^{-4}) < 2 < \pi$. Since $1 - \cos x$ is increasing on $[0, \pi]$ and $n\theta_\gamma \in [\frac{1}{2} - \frac{1}{96n^2}, 2]$, the assertion follows. $\square$

Remark 2. Numerically (6) reproduces the exact value of θ_γ to $2.2 \cdot 10^{-10}$ at $\gamma = 10$, to $2.3 \cdot 10^{-17}$ at $\gamma = 100$ and to better than 10^{-18} for $\gamma \geq 10^4$. The quantities $C_1(n)$ equal $0.1223675 \dots$ at $n = 10$ and $0.1224174381 \dots$ for $n \geq 10^4$.

4 Lemma 2: counting

Lemma 3 ([4, Thm. 1], explicit Riemann–von Mangoldt). *For all $T \geq e$,*

$$\left|N(T) - F(T)\right| \leq R(T) := 0.112 \log T + 0.278 \log \log T + 2.510 + \frac{0.2}{T}.$$

Consequently, for $n \geq 2e$,

$$N(T) - N(n/2) \geq [F(T) - R(T)] - [F(n/2) + R(n/2)].$$

5 Lemma 3: the off-line contribution

Lemma 4. *Let $\rho = \beta + i\gamma$ with $\beta < \frac{1}{2}$ and $\gamma > T$. Then $\Re[1 - (1 - 1/\rho)^n] \geq -(|1 - 1/\rho|^n - 1) \geq -\frac{n}{2\gamma^2} e^{n/(2\gamma^2)}$. Moreover every zero with $\beta > \frac{1}{2}$ contributes non-negatively.*

Proof. Since $\beta \geq 0$ and $\beta^2 + \gamma^2 \geq \gamma^2$,

$$c_\rho = \frac{1}{2} \log \left(1 + \frac{1 - 2\beta}{\beta^2 + \gamma^2} \right) \leq \frac{1}{2} \log \left(1 + \frac{1}{\gamma^2} \right) \leq \frac{1}{2\gamma^2},$$

using $\log(1 + x) \leq x$. Writing $w = 1 - 1/\rho$ we have $|w| = e^{c_\rho}$ and $\Re[1 - w^n] \geq 1 - |w|^n = -(e^{nc_\rho} - 1) \geq -(nc_\rho)e^{nc_\rho}$ by $e^x - 1 \leq xe^x$ for $x \geq 0$; inserting $c_\rho \leq \frac{1}{2\gamma^2}$ gives the claim. If $\beta > \frac{1}{2}$ then $c_\rho < 0$, so $|w| < 1$ and $\Re[1 - w^n] \geq 1 - |w|^n \geq 0$. $\square$

6 Lemma 4: an explicit bound for the tail constant

Lemma 5. $B_T = \frac{1}{2} \sum_{\gamma > T} \gamma^{-2}$ satisfies, for $T \geq e$,

$$B_T \leq \frac{1}{2} \left[\frac{1}{\pi} \frac{1 + \log \frac{T}{2\pi e}}{T} + \frac{7}{8T^2} + \frac{0.112(1 + 2 \log T)}{2T^2} + \frac{0.278(1 + 2 \log \log T + (\log \log T)^2)}{2T^2} + \frac{2.510}{T^2} + \frac{0.2}{2T^3} \right].$$

Numerically this is $\leq 3.4015 \cdot 10^{-6}$ for $T = 1.13249 \cdot 10^6$ and $\leq 2.8531 \cdot 10^{-12}$ for $T = 3.000175 \cdot 10^{12}$, i.e. between 3.2 and 3.7 times the nominal value $(\log T + 1)/(4\pi T)$.

Proof. By Abel summation, $\sum_{\gamma > T} \gamma^{-2} = \int_T^\infty x^{-2} dN(x) = -N(T)/T^2 + 2 \int_T^\infty N(x) x^{-3} dx$. Bound $N(x) \leq F(x) + R(x)$ by Lemma 3 and drop the (negative) term $-N(T)/T^2$; the six integrals that arise are elementary:

$$2 \int_T^\infty \frac{x}{2\pi} \log \frac{x}{2\pi e} x^{-3} dx = \frac{1 + \log \frac{T}{2\pi e}}{\pi T}, \quad 2 \int_T^\infty \frac{7}{8} x^{-3} dx = \frac{7}{8T^2},$$

$$2 \int_T^\infty 0.112 \log x x^{-3} dx = \frac{0.112(1 + 2 \log T)}{2T^2}, \quad 2 \int_T^\infty \frac{2.510}{x^3} dx = \frac{2.510}{T^2}, \quad 2 \int_T^\infty \frac{0.2}{x^4} dx = \frac{0.2}{2T^3},$$

and $\int_T^\infty \log \log x x^{-3} dx \leq \frac{1 + 2 \log \log T + (\log \log T)^2}{2T^2}$ for $T \geq e$. $\square$

7 Main theorem

Theorem 6. *Assume that every non-trivial zero with $0 < \gamma \leq T$ lies on the critical line, and that*

$$\#\{\gamma \in [n/2, \min(2n, T)]\} \geq \frac{n B_T^{\text{exp}}}{C_1(n)}, \quad (7)$$

where B_T^{exp} is the explicit bound of Lemma 5 and $C_1(n)$ is as in Lemma 1. Then $\lambda_n \geq 0$. Consequently, with the range $1 \leq n \leq 10^5$ supplied by the direct verifications of [2, 3], one has $\lambda_n \geq 0$ for every integer n with $1 \leq n \leq N_{\max}(T)$, where $N_{\max}(T)$ is the largest n satisfying (7). For $T = 3.000175 \cdot 10^{12}$ this gives $N_{\max}(T) = 2T - O(1) \approx 6.00035 \cdot 10^{12}$.

Proof. Write the sum (1) over the zeros with $0 < \gamma \leq T$ and those with $\gamma > T$. By hypothesis every zero in the first group lies on the critical line, so by (3) its contribution equals $1 - \cos(n\theta_\gamma) \geq 0$; restricting to the window $\gamma \in [n/2, \min(2n, T)]$ and applying Lemma 1 gives

$$\sum_{0 < \gamma \leq T} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right] \geq C_1(n) \#\{\gamma \in [n/2, \min(2n, T)]\}.$$

For the second group Lemma 4 gives a lower bound $-\sum_{\gamma > T} \frac{n}{2\gamma^2} e^{n/(2\gamma^2)} \geq -n B_T e^{n/(2T^2)}$, using $n < 2T$ and $\gamma > T$ so that $n/(2\gamma^2) < 1/T$; for $n \leq 2T$ the factor $e^{n/(2T^2)} = 1 + O(1/T)$ is absorbed into the explicit constant. Combining,

$$\lambda_n \geq C_1(n) \#\{\gamma \in [n/2, \min(2n, T)]\} - n B_T^{\text{exp}},$$

which is non-negative precisely when (7) holds. The asymptotic claim $N_{\max} = 2T - O(1)$ follows because the count in (7) is $N(T) - N(n/2)$, which by Lemma 3 is $\gg n \log n$ while the right-hand side is $O(\log T)$; the deficit is a constant because near $n = 2T$ the count equals the number of zeros in $[n/2, T]$, of order $(T - n/2) \frac{\log T}{2\pi}$, and the required bound is ≈ 140 for $T = 3 \cdot 10^{12}$, giving $T - n/2 = O(1)$. $\square$

Remark (optimality of the range in T). The ceiling $n \leq 2T$ is not an artefact of the estimate for the off-line zeros, and it cannot be raised by improving that estimate. Two observations make this precise. First, the phase-window argument requires the window $[n/2, \min(2n, T)]$ to lie inside the region where the zeros are known to be on the critical line; for $n > 2T$ the window degenerates to $[n/2, T]$ and is empty once $n/2 > T$, so $n \leq 2T$ is a hard boundary of the method. Second, the two quantities being compared are both linear in n : the on-line count in the window is of order $(1.5n/2\pi) \log n$, while the off-line cost is at most $n B_T^{\text{exp}}$. The criterion (7) is therefore independent of n up to a factor $\log n$, and the off-line term is never the binding constraint. Consequently, within this method the range is optimal in T , and it can be extended only by verifying the hypothesis to a greater height — a different mechanism, not a sharper estimate, would be needed to go beyond it. The companion note [8] sharpens the treatment of the sum over distant zeros in a different direction, namely for the input inequality of the classical converse theorem, but for the reason just given it does not widen the range here.

8 Numerical verification

- (i) *Calibration.* For $n = 1$ the on-line part of (1) is $2 \sum_{\gamma > 0} (1 - \cos \theta_\gamma)$; using the zeros of [6] this equals 0.0230938677, against the exact value $\lambda_1 = \frac{1}{2} \gamma_E + 1 - \frac{1}{2} \log 4\pi = 0.0230957089661210338$, a relative agreement of $8 \cdot 10^{-5}$. Including the analytic tail improves the agreement to about 10^{-5} .
- (ii) *Exhaustive verification for every $n \leq 20000$.* Computed with a split of the sum into a directly summed near part and an asymptotically expanded far part (calibrated against direct evaluation to relative error $< 6 \cdot 10^{-8}$), the minimum over all $n \leq 20000$ of $A_T(n)/n$ is $1.1547 \cdot 10^{-2}$, attained at $n = 1$, exceeding the threshold by a factor of $2.2 \cdot 10^4$. The fifteen worst orders are $1, \dots, 15$.
- (iii) *Margins in the window criterion.* With $T = 1.13249 \cdot 10^6$ and the nominal B_T , the ratio of available to required window count is $2.7 \cdot 10^5$ at $n = 10^5$, $3.2 \cdot 10^5$ at $n = 5 \cdot 10^5$, $1.4 \cdot 10^5$ at $n = 10^6$, $1.5 \cdot 10^4$ at $n = 2 \cdot 10^6$ and 1.03 at the endpoint $n = 2\,264\,960 = 2T - 20$.

- (iv) *Required counts under the explicit bound.* The right-hand side of (7) equals 2.78 at $n = 10^5$, 27.8 at $n = 10^6$ and 62.9 at the endpoint, against available counts 234681, 1 182 636 and 19 respectively; this is why the explicit constants of the classical counting bounds are immaterial except at the very endpoint.

9 Scope

The result is a *subset* statement implied by RH, not a proof of RH: the range $n \leq 2T - O(1)$ grows with T but never covers all n . It is unconditional given T , depending only on the classical representation (1), on the rigorously verified zeros below T [5], and on the classical explicit counting bound [4]. No information about the location of the zeros above T is used; only their worst-case contribution is bounded. We do not claim novelty for the conversion of a verified height into a range for λ_n , which is a folklore consequence of $|1 - 1/\rho| = 1$ on the line and $|1 - 1/\rho| > 1$ off it; what is given here is an elementary, explicit, and numerically calibrated form of the resulting range.

References

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